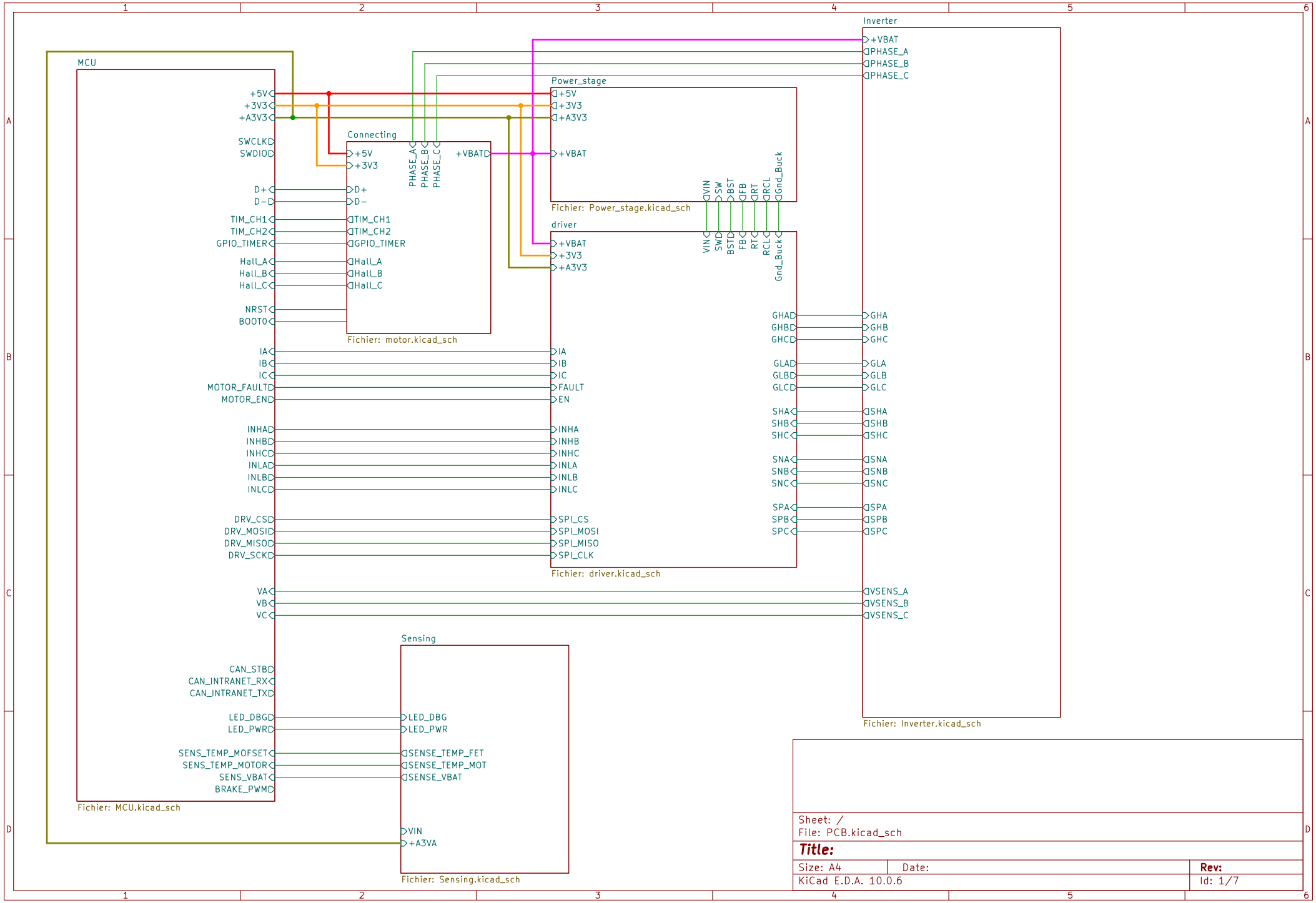
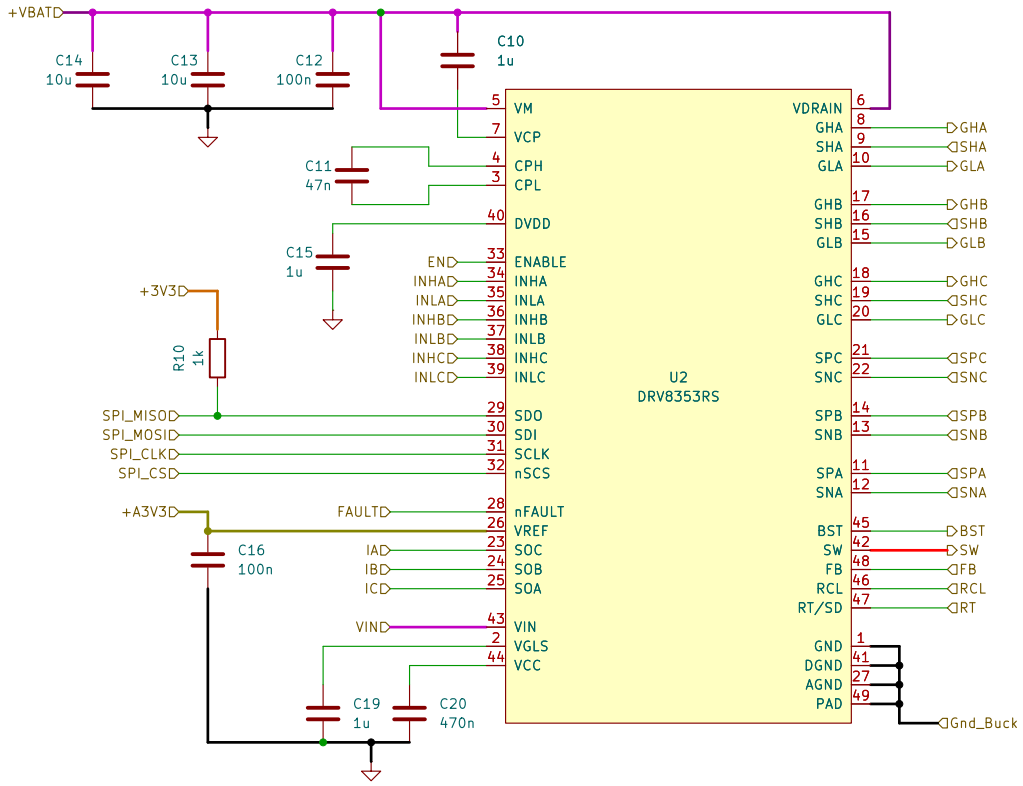
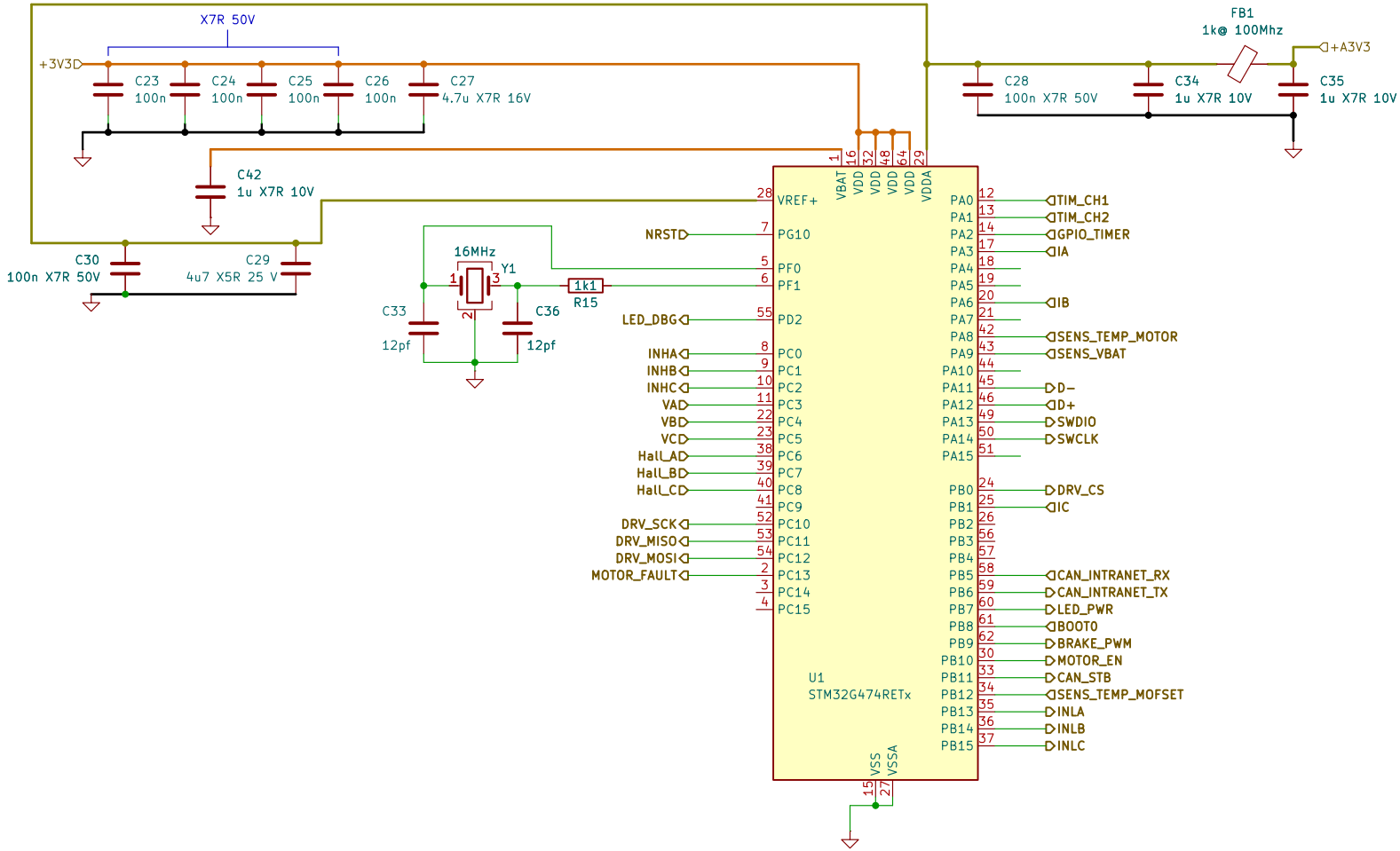

MCU
+5V
+3V3
+A3V3
SWCLK
SWDIO
D+
D-
TIM_CH1
TIM_CH2
GPIO_TIMER
Hall_A
Hall_B
Hall_C
NRST
BOOT0
IA
IB
IC
MOTOR_FAULT
MOTOR_EN
INHAD
INHBD
INHCD
INLAD
INLBD
INLCD
DRV_CSD
DRV_MOSI
DRV_MISO
DRV_SCK
VA
VB
VC
CAN_STBD
CAN_INTRANET_RX
CAN_INTRANET_TX
LED_DBGD
LED_PWRD
SENS_TEMP_MOFSET
SENS_TEMP_MOTOR
SENS_VBAT
BRAKE_PWM
Fichier: MCU.kicad_sch
Connecting
+5V
+3V3
PHASE_A
PHASE_B
PHASE_C
+VBATD
D+
D-
TIM_CH1
TIM_CH2
GPIO_TIMER
Hall_A
Hall_B
Hall_C
NRST
BOOT0
Fichier: motor.kicad_sch
Power_stage
+5V
+3V3
+A3V3
+VBAT
VIN
SW
BST
FB
RT
RCL
Gnd_Buck
Fichier: Power_stage.kicad_sch
driver
+VBAT
+3V3
+A3V3
VIN
SW
BSTD
FB
RT
RCL
Gnd_Buck
IA
IB
IC
FAULT
EN
INHA
INHB
INHC
INLA
INLB
INLC
SPI_CS
SPI_MOSI
SPI_MISO
SPI_CLK
GHAD
GHBD
GHCD
GLAD
GLBD
GLCD
SHA
SHB
SHC
SNA
SNB
SNC
SPA
SPB
SPC
Fichier: driver.kicad_sch
Inverter
+VBAT
PHASE_A
PHASE_B
PHASE_C
GHA
GHB
GHC
GLA
GLB
GLC
SHA
SHB
SHC
SNA
SNB
SNC
SPA
SPB
SPC
VSENS_A
VSENS_B
VSENS_C
Fichier: Inverter.kicad_sch
Sensing
LED_DBG
LED_PWR
SENSE_TEMP_FET
SENSE_TEMP_MOT
SENSE_VBAT
VIN
+A3VA
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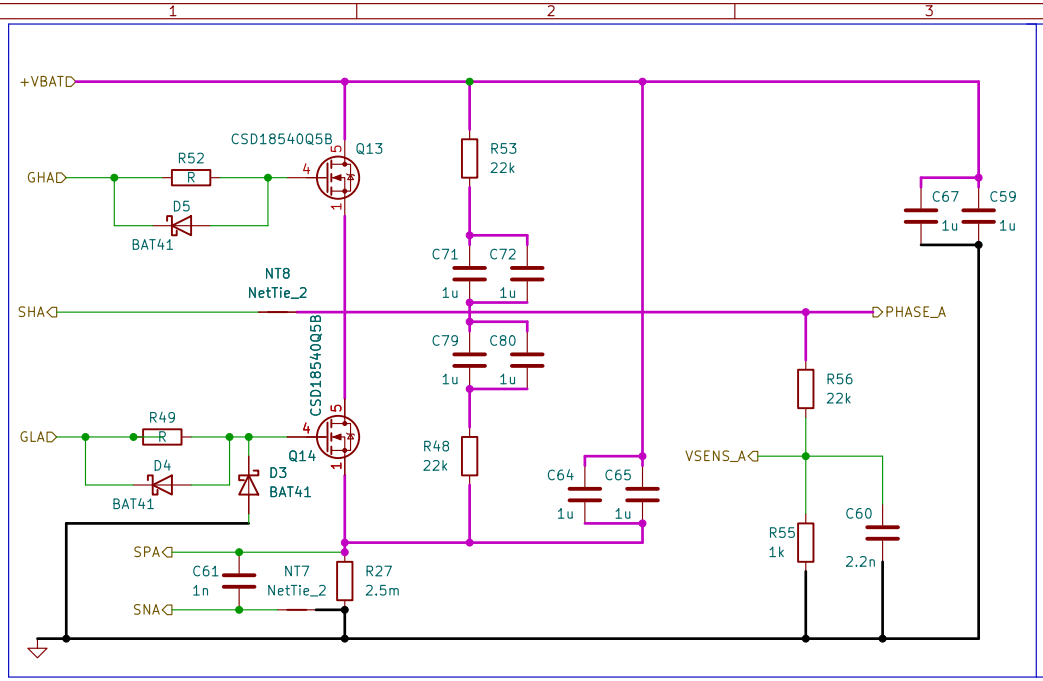
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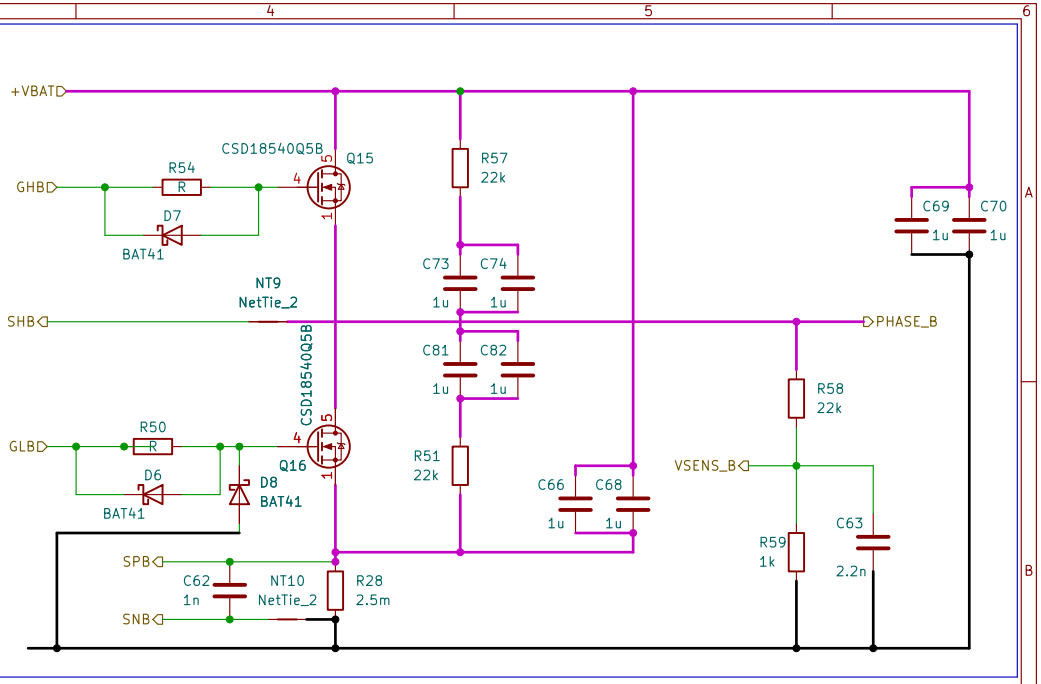


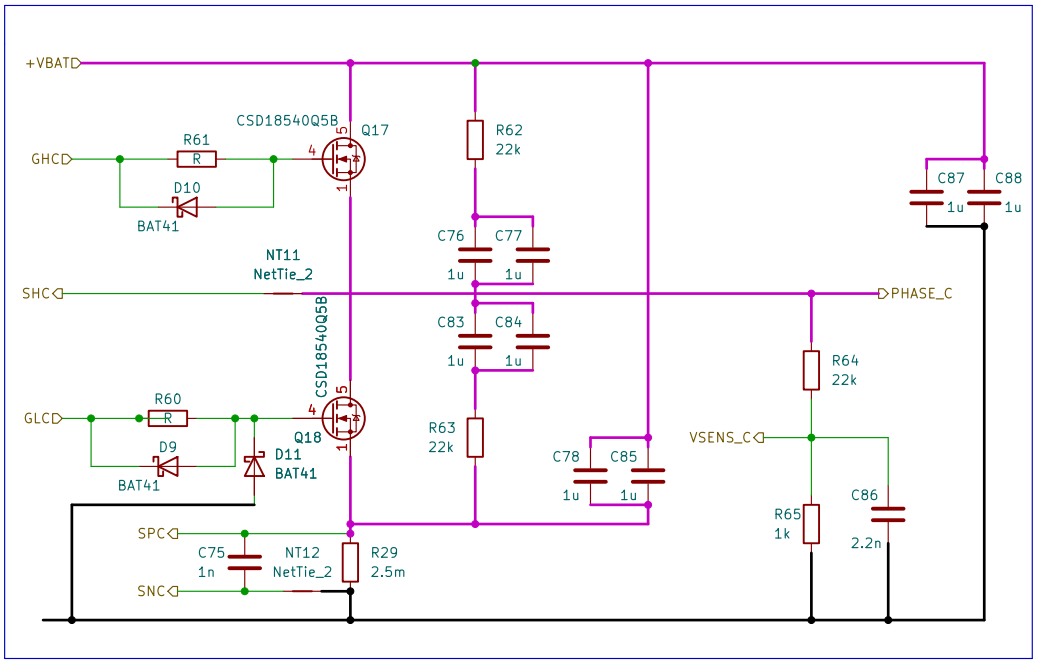
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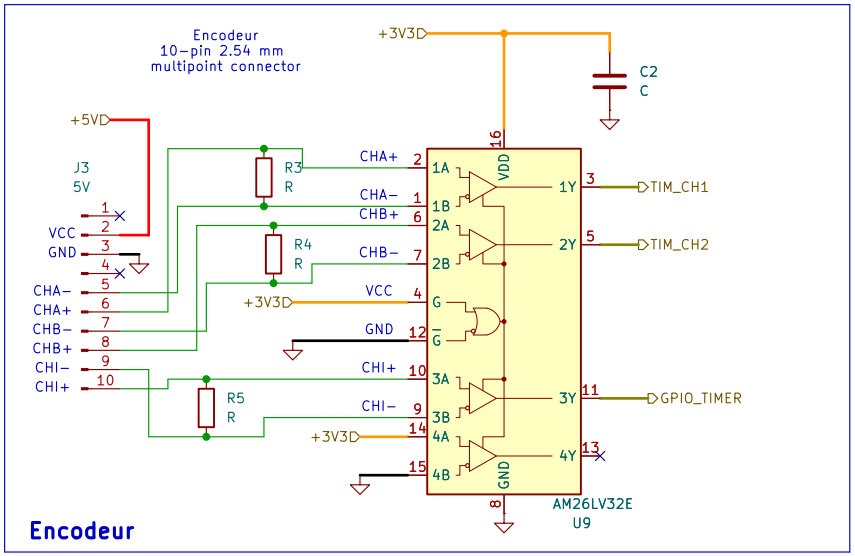
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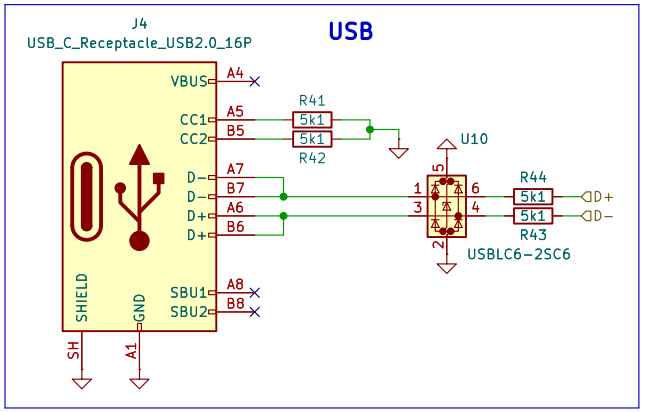


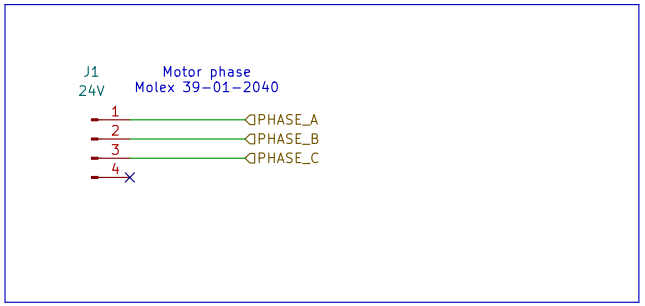


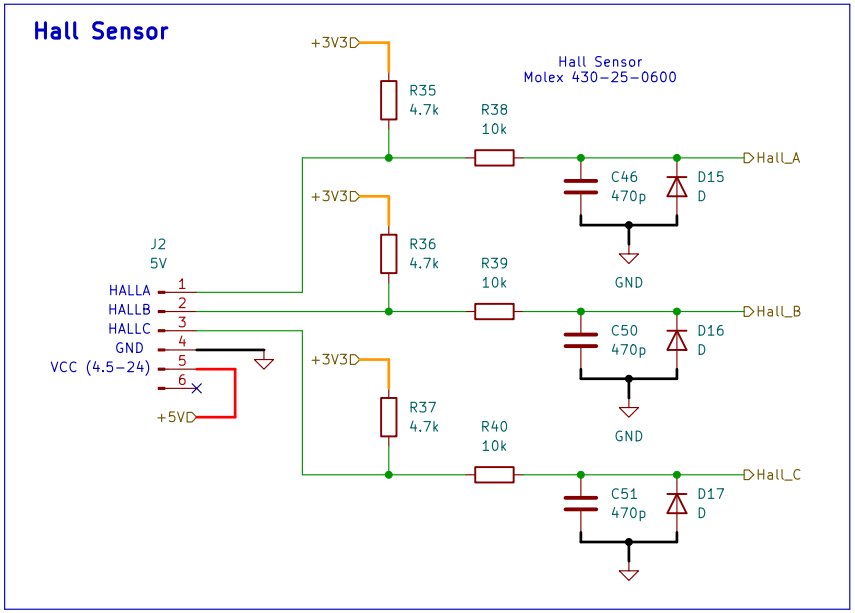


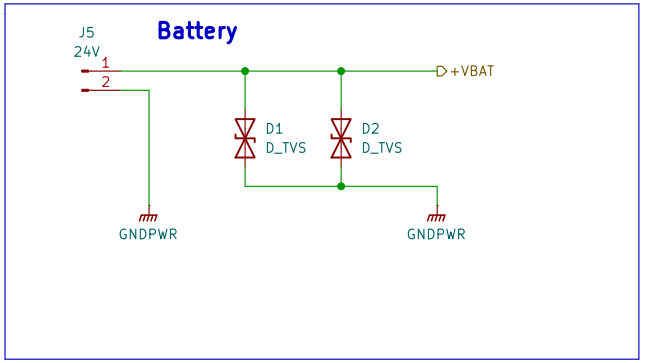
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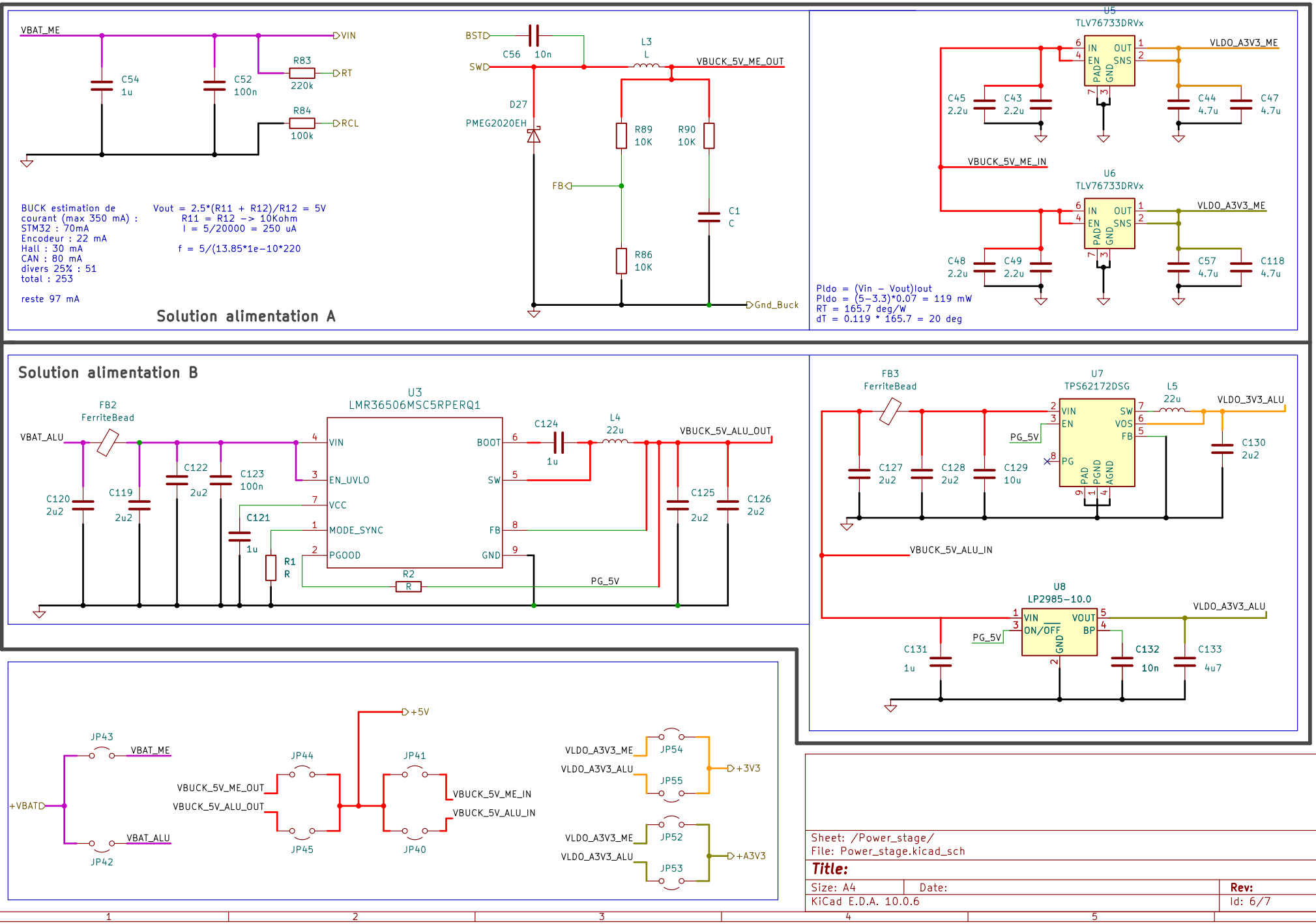


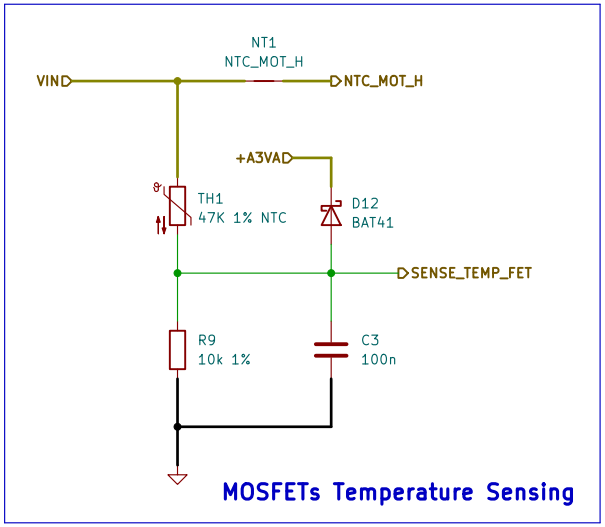


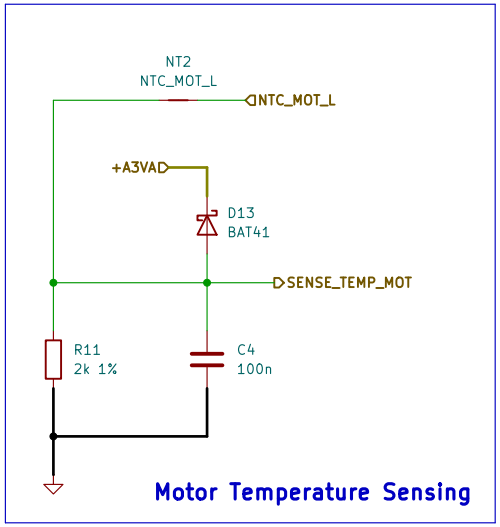
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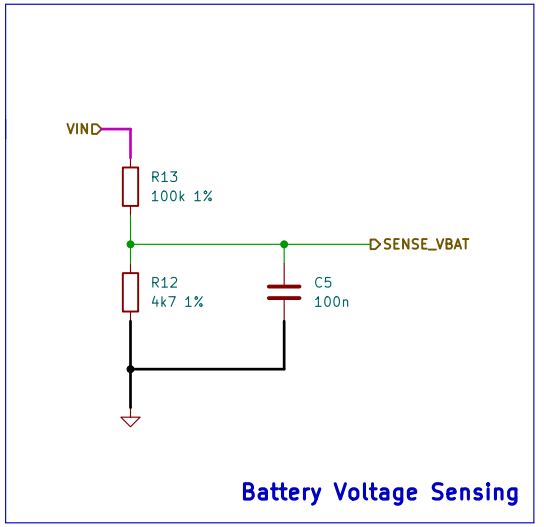
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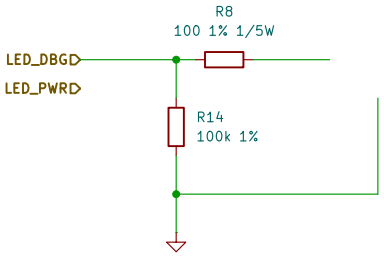
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VBAT_ME
VIN
C54
1u
C52
100n
R83
220k
RT
R84
100k
RCL
BUCK estimation de courant (max 350 mA) :
STM32 : 70mA
Encoder : 22 mA
Hall : 30 mA
CAN : 80 mA
divers 25% : 51
total : 253
reste 97 mA
Vout = 2.5*(R11 + R12)/R12 = 5V
R11 = R12 -> 10Kohm
I = 5/20000 = 250 uA
f = 5/(13.85*1e-10*220
Solution alimentation A
BSTD
SW
C56 10n
L3
L
VBUCK_5V_ME_OUT
D27
PMEG2020EH
R89
10K
R90
10K
FB
C1
C
R86
10K
Gnd_Buck
U5
TLV76733DRVx
IN
OUT
EN
SNS
PAD
GND
VLDO_A3V3_ME
C45
2.2u
C43
2.2u
C44
4.7u
C47
4.7u
VBUCK_5V_ME_IN
U6
TLV76733DRVx
C48
2.2u
C49
2.2u
C57
4.7u
C118
4.7u
Pldo = (Vin – Vout)Iout
Pldo = (5–3.3)*0.07 = 119 mW
RT = 165.7 deg/W
dT = 0.119 * 165.7 = 20 deg
Solution alimentation B
U3
LMR36506MSC5RPERQ1
FB2
FerriteBead
VBAT_ALU
VIN
EN_UVLO
VCC
MODE_SYNC
PGOOD
BOOT
SW
FB
GND
C120
2u2
C119
2u2
C122
2u2
C123
100n
C121
1u
R1
R
R2
R
C124
1u
L4
22u
VBUCK_5V_ALU_OUT
C125
2u2
C126
2u2
PG_5V
FB3
FerriteBead
U7
TPS62172DSG
VIN
EN
SW
VOS
FB
PG
PAD
PGND
AGND
L5
22u
VLDO_3V3_ALU
C130
2u2
C127
2u2
C128
2u2
C129
10u
PG_5V
VBUCK_5V_ALU_IN
U8
LP2985-10.0
VIN
ON/OFF
GND
VOUT
BP
VLDO_A3V3_ALU
C131
1u
C132
10n
C133
4u7
JP43
VBAT_ME
+VBATD
VBAT_ALU
JP42
+5V
JP44
JP45
VBUCK_5V_ME_OUT
VBUCK_5V_ALU_OUT
JP41
JP40
VBUCK_5V_ME_IN
VBUCK_5V_ALU_IN
VLDO_A3V3_ME
VLDO_A3V3_ALU
JP54
JP55
+3V3
JP52
JP53
+A3V3
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File: Power_stage.kicad_sch
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